

QUICKTAP

POS Payment App - Design & Flow Document

UK Small Merchants - February 2026

What this document covers

A clickable POS prototype for UK small merchants. Card and QR payment flows, transaction history, and in-app refunds. UI and simulated logic only - no real payments, no backend, no integrations.

Sections: Product Overview · Target Users · Design Principles · Competitive Context · Payment Flow · Screen Inventory · Error Design · Scope & Positioning

1 - Product Overview

QuickTap is designed to feel calm under pressure.

Perceived speed and error confidence - not feature depth.

A clickable prototype demonstrating the full payment journey for card and QR. Every screen state - waiting, processing, approved, failed - is covered. Transaction history and in-app refunds are included. Nothing else.

In scope	Out of scope
Card payment - all states (waiting, processing, approved, failed)	Real payment processing - no API calls, no card network
QR payment with countdown timer and auto-refresh	Product catalogue - amount entry only
Transaction history (today default, Today/Week toggle)	Cash, voucher, split pay
In-app refund - 3 taps, confirmation modal	Staff accounts, settings, analytics

2 - Target Users

"I need to take a payment right now. Three people behind this customer. I cannot read small text. I cannot guess what to do next."

Market traders, café owners, mobile service providers. Always doing two things at once. The app has to work without their full attention, at arm's length, in any lighting condition.

User need	Design response
Payment in under 10 seconds from home screen	One primary action on home: Charge. Four taps to money.
Know instantly if a payment worked	Full-screen colour confirmation - green or red, unmissable.
Know what to do when something goes wrong	Every error names the cause and gives the cashier a sentence to say out loud.
See today's total without navigating	Daily total on the home screen, visible the moment the app opens.
Issue a refund without opening a laptop	Refund surfaces on the transaction detail screen - 3 taps from history.

3 - Design Principles & Decisions

Principle	Rule	Why
Big & Bold	Primary text $\geq 32\text{px}$. Buttons $\geq 60\%$ screen width.	Merchants read at arm's length in sunlight with one eye on a queue. Small text fails this context.
Minimal Steps	Card or QR payment: four taps max.	Speed is the product. More than four taps and the flow has failed the merchant.
State is King	One state per screen. Full-screen colour + icon + plain-English headline.	A cashier should never have to read three words to know where they are. Colour registers before text.
Recovery Design	Every failure answers: What happened? What do I tell the customer? What next?	A generic "Error - try again" is useless when a cashier is standing in front of a waiting customer.

Key design decisions

Decision	Rejected alternative	Reason
Amount entry first - no catalogue	Product tile grid	Small merchants know their prices. A catalogue adds setup, maintenance, and taps for zero benefit here.
Two payment methods only	Full tender list	94.6% of UK transactions are contactless card. QR is growing. Every other method is noise at this stage.
QR equal weight to Card	QR in "More options" submenu	Burying QR kills adoption. Lower QR fees benefit merchants - equal placement nudges the right behaviour.
Full-screen colour states	Toast or banner notifications	Full-screen colour is visible peripherally. A banner requires the cashier to look directly at the phone.
Processing screen locked - no buttons	Cancel always visible	Tapping Cancel mid-transaction causes double-charge anxiety. Lock it. Show movement. Wait 3 seconds.
Success auto-dismisses at 4 seconds	Manual dismiss only	In a queue, the cashier should tap nothing after success. 4 seconds registers; short enough to keep moving.
In-app refund - 3 taps	Web back-office only	A cashier cannot open a laptop mid-dispute. In-app refund is one of the stickiest features in the product.
History defaults to Today	All-time or last 30 days	Today is always the relevant window. All-time first creates cognitive load with no benefit.

4 - Design Context & Competitive Observations

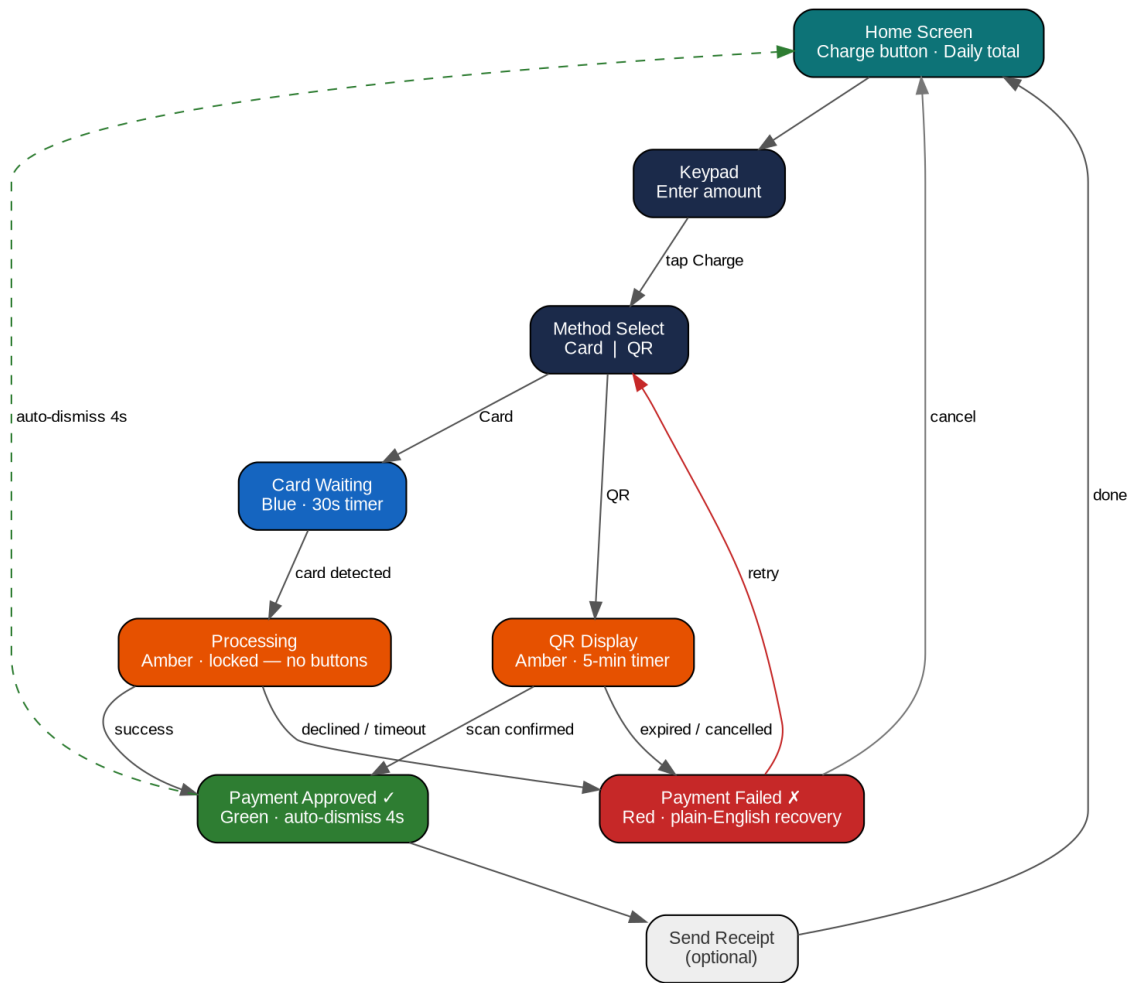
Observations from using SumUp, Square, and Zettle in practice with UK small merchants. Not a feature comparison - these are the friction points that directly shaped QuickTap's design.

Observation	What SumUp / Square / Zettle do	QuickTap response
Processing state is ambiguous	A spinner with no countdown. Merchants watch anxiously, unable to tell if the card failed or the app froze.	Full-screen amber + pulsing animation. Cancel option at 15 seconds. The merchant always knows something is happening.
Error messages don't help	"Transaction failed", "Declined", "Timed out" - merchant guesses the cause and re-enters the amount with the customer still waiting.	Every error names the specific cause and gives the cashier a sentence to say out loud. No codes, no re-entry from scratch.
Transaction history requires navigation	Leave the payment flow, open Sales/Transactions, scroll to the last entry. Too slow under queue pressure.	History is one tap from home, always showing today with a running total. Under 3 seconds from home to confirmation.
Refunds are fiddly mid-dispute	Technically in-app, but locating the transaction and confirming amount and reason adds friction exactly when speed matters.	Refund button on the transaction detail screen - 3 taps, no reason field, no ambiguity. Failed transactions hide the button entirely.
Double-charge anxiety	Poor connectivity + retry = two successful charges. Merchants manually check history and issue corrective refunds.	Processing screen is locked - no retry while in flight. "Card tapped twice" error confirms no charge and prompts a clean restart.

5 - Core Payment Flow

Flow	Steps
Payment	Home → Charge → Keypad → Method Select → Card Waiting / QR Display → Processing → Approved ✓ or Failed X → (Receipt) → Home
Refund	Home → History → Transaction List → Transaction Detail → Refund → Confirm → Done

Node colours map directly to the state system: Teal = navigation, Blue = waiting for customer, Amber = active/processing, Green = success, Red = failure.



State colour system

State	Colour	Screen(s)	What it signals
Waiting	Blue	Card Waiting, QR Display	Calm. Ready. Waiting for customer input - no urgency.
Processing	Amber	Card Processing	Active. Do not interrupt. Something is happening right now.
Approved	Green	Payment Approved	Universal success. Full-screen - visible from any angle.
Failed	Red	All error states	Universal failure. Cashier knows before reading a word.

6 - Screen Inventory

Screen	Purpose	Key elements	Design note
Home	Starting point. One tap to first payment.	Today's total (large). Transaction count. Last 3 entries. CHARGE (primary). History icon. Connection status.	No competitor shows daily takings on open. Answers the merchant's first question before they touch Charge.

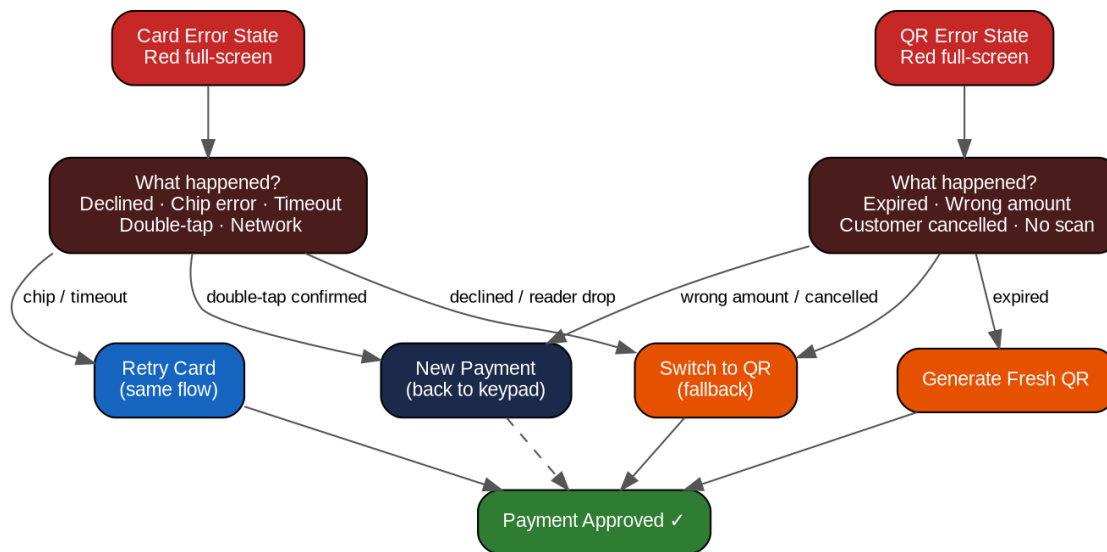
Screen	Purpose	Key elements	Design note
Keypad	Amount entry. Nothing else.	Full-width keypad. Large digit display. Backspace. Decimal. CHARGE button. No catalogue.	One thumb, in a glove, in sunlight. Catalogue adds taps and setup overhead that don't belong here.
Method Select	Card or QR - two options only.	Amount confirmed (large). CARD and QR CODE at equal visual weight. Back arrow.	94.6% contactless card, QR growing. Equal weight matters - burying QR kills adoption and QR fees benefit merchants.
Card Waiting	Customer taps, inserts, or swipes.	Blue screen. Animated contactless icon. Amount. "Tap, insert or swipe." 30s countdown. Cancel (bottom).	Cashier turns the phone - no verbal instruction needed. Timer prevents frozen-screen panic.
Processing	Transaction in flight. Lock the screen.	Amber screen. Pulsing animation. Amount shown. No buttons.	Tapping Cancel mid-transaction causes worse problems than waiting 3 seconds. Lock it. Show movement.
QR Display	Customer scans with their banking app.	QR code (maximum size). Amount above QR. "Scan with your banking app." Amber border. 5-min timer. Auto-refresh.	Largest possible QR - small codes fail in dim light. Amount shown above so customer verifies before scanning.
Approved	Payment confirmed. Unmissable.	Full green screen. Animated checkmark. "Payment Approved" 36px+. Amount. Time. New Payment (primary). Send Receipt. Auto-dismiss 4s.	Emotional payoff of the flow. Auto-dismiss = zero extra taps after success in a busy queue.
History	All today's transactions. One tap from home.	Today's list by default. Amount + method + time + status badge per row. Today/Week toggle. Running total at top.	Competitors bury this. One tap, defaulting to today. Colour-coded badges make the list scannable in 2 seconds.
Transaction Detail + Refund	Full details. Refund in 2 taps.	Amount (large). Date/time/method. Status badge. Card last 4 / "QR Payment". REFUND (shown only on successful transactions). Confirm modal.	In-app refund in front of a customer is a confidence moment. Failed transactions never show the Refund button.

7 - Error & Recovery Design

Every error screen answers three questions: what happened, what should I tell the customer, what do I tap next. No codes, no guessing, no re-entering the amount from scratch.

Error screen anatomy	
Background	Red - full-screen, unmistakable before reading a word
Headline	What happened - one plain-English sentence
Sub-text	What to tell the customer - something they can say out loud right now
Primary button	Retry or most relevant recovery action
Secondary button	Cancel / New Payment

Error recovery flow



Card error states

Trigger	✗ Competitor	✓ QuickTap message	Next action
Card declined	"Payment Failed. Try again."	"Card declined by bank. Ask customer to try contactless or use a different card."	Retry or switch to QR
Chip read error	"Error reading card."	"Card chip couldn't be read. Ask customer to tap instead of inserting."	Retry with tap
No card in 30s	"Transaction cancelled."	"No card detected in 30 seconds. Ready for a new attempt when you are."	Restart payment
Processing timeout	"Something went wrong."	"Connection took too long. Check your Wi-Fi or signal, then retry."	Check signal + retry
Double-tap	"Error."	"Card tapped twice. This transaction was not charged. Start a fresh payment."	New payment

QR error states

Trigger	✗ Competitor	✓ QuickTap message	Next action
QR expired (5 min)	"QR expired. Generate new."	"QR timed out - no charge made. Tap to generate a fresh code."	Generate new QR
Wrong amount paid	"Payment error."	"Customer paid a different amount. Transaction not accepted. Show QR again."	Show QR again
Customer cancelled	"Transaction cancelled."	"Customer cancelled. No charge made. Ready when they are."	Retry or switch to card
No scan in 5 min	"Session timed out."	"No payment received - QR expired. Switch to card or retry."	Switch or retry

8 - Scope Boundaries & Final Positioning

Prototype scope

Feature	Status	Notes
Home screen with daily total	☑	Simulated - increments as payments complete in session
Keypad / amount entry	☑	Fully interactive, clear and backspace
Method selection (Card / QR)	☑	Both fully routed
Card payment - all states	☑	Waiting → Processing → Approved / Failed
QR payment with countdown	☑	Static QR, simulated confirm after timer
Payment Approved full-screen	☑	Green screen, animation, auto-dismiss 4s
Card declined error screen	☑	Plain-English recovery + Retry
QR expired error screen	☑	Regenerate option shown
Transaction history list	☑	Simulated data, Today/Week toggle, colour badges
Transaction detail + Refund	☑	Confirmation modal included
Receipt sending (SMS/Email)	○	UI only - buttons shown, no actual send
Real payment processing	✗	All states simulated - no API calls
Product catalogue	✗	Out of scope - amount entry only
Multi-staff accounts / settings	✗	Single merchant view only

Simulated behaviour

Behaviour	Logic
Card payment	2-second processing delay → Approved (90%) or Declined (10%)
QR payment	5-minute countdown → auto-resolves to Approved to show success flow
Transaction history	Pre-populated with realistic mixed data - card and QR, approved and failed
Daily total	Increments in session as prototype payments complete

QuickTap is designed to feel reliable during peak-hour pressure.

Calmer. Faster. More reassuring. The same core flow - done right.